

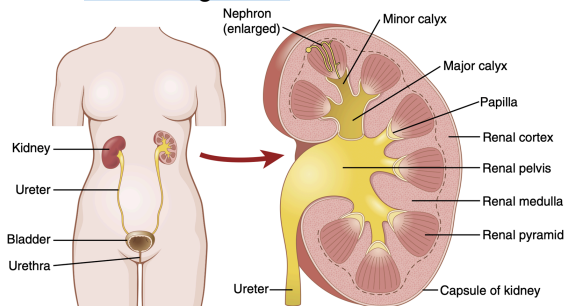
INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #4: RENAL SYSTEM | OPHT 1-1 | MIDTERMS

By Accel Jimenez

HOMEOSTATIC FUNCTIONS OF KIDNEYS

- Excretion of metabolic waste products and foreign chemicals
- Regulation of water and electrolyte balances
- Regulation of body fluid osmolality and electrolyte concentrations
- Regulation of arterial pressure
- Regulation of acid- base balance
- Regulation of erythrocyte production
- Secretion, metabolism, and excretion of hormones
- Gluconeogenesis



External Anatomy: Kidneys

- Renal capsule
 - Barrier to trauma
- Adipose tissue
- Renal fascia
 - Anchors the kidneys to the abdominal wall
- Renal cortex
 - Outer region
- Renal medulla
 - Inner region
- (+) renal pyramids

Renal Blood Supply

- Blood flow to the two kidneys is normally about 22 percent of the cardiac output, or 1100 ml/min
- INTERLOBULAR AA aka radial arteries
- Two capillary beds: the glomerular and peritubular capillaries
- High hydrostatic pressure in the glomerular capillaries (about 60 mm Hg)
- peritubular capillaries (about 13 mm Hg)
- rapid fluid filtration
- rapid fluid reabsorption

Nephron

- Structural and Functional unit of the kidneys
- There are two major types of nephrons:
 - Cortical Nephron
 - Juxtamedullary Nephron
- Two major parts of nephron:
 - **Renal Corpuscle:** Afferent arterioles, glomerular capillaries, efferent arterioles, podocytes, mesangial cells, JG cells, Bowman's capsule & space
 - **Renal Tubules:** Proximal Convoluted Tubules (PCT), Loop of Henle, Distal Convoluted Tubule (DCT), Collecting Duct (CD)

Endocrine Function of Kidneys

Erythropoietin (EPO)	• Controls the rate of RBC production • Released in response to hypoxia → stimulates RBC proliferation in bone marrow • Chronic Kidney Disease (CKD) causes decrease EPO → chronic anemia
Calciferol (Vitamin D)	• PCT converts 25-OH vitamin D3 → 1,25-(OH)2 vitamin D3 (Calcitriol, active form) • Conversion is mediated by 1α-hydroxylase (which is upregulated by PTH)
Prostaglandins	• Effects of prostaglandins (PGE2 and PGI2) on the kidneys: increased renal blood flow and GFR • Paracrine secretion vasodilates the afferent arterioles → increase renal blood flow • NSAIDs block renal protective prostaglandin synthesis → decrease GFR → Acute Kidney Injury
Dopamine	• Forms in PCT cells • Promotes natriuresis • At low doses: dilates interlobular arteries, afferent arterioles, efferent arterioles → inc. renal blood flow • At high doses: acts as a vasoconstrictor

Urine Formation

Glomerular filtration

- (+) Filtration membrane – allows water, small solutes to pass through(–) proteins, platelets, blood cells
- **Net filtration pressure** – 10 mm Hg
- **Glomerular filtration rate (GFR)** – amount of filtrate formed in all renal corpuscle (Glomerulus + Bowman's capsule)
- **125 mL/min** – males
- **105 mL/min** – females

Tubular Reabsorption

- Proximal convoluted tubule (workhorse of the nephron)
 - **65% reabsorption of water**
 - **Concentrated**
 - 65% of most filtered electrolytes is reabsorbed in the proximal tubule.
 - **Thin descending Loop of Henle**
 - Reabsorption of **water**
 - Concentrated (> PCT)

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- **Thick ascending Loop of Henle**
 - Reabsorption of solutes (Sodium, chloride, potassium and other ions)
 - More dilute
- **Distal convoluted tubule and connecting tubule**
 - Diluted
 - Pale yellow to orange (color caused by urochrome / urobilin)
 - ADH / Vasopressin attached to DCT
- **Medullary Collecting Duct**
 - Reabsorb less than 5% of the filtered water and sodium
 - **final site for processing the urine**
 - play a critical role in determining the final urine output of water and solutes

Secretion

- movement from peritubular capillaries to interstitium to Tubules
- Excretion = (Amount Filtered) – (Amount Reabsorbed) + (Amount Secreted)
- Excretion is NOT synonymous with secretion.
- Normal urine volume of the bladder after urination of normal young adult : <50mL

Glomerular Filtration Rate

- Amount filtered in the glomerular capillaries per unit time
- Normal value: 125mL/min or 180L/day
 - Determined by Starling forces at the level of the glomerular capillary (glomerulus)
- **GFR males vs. females: 20% higher in males**
- **BUN and Creatinine increase when GFR decreases**
 - GFR decreases with age, but **Creatinine remains constant** due to decreased muscle mass

Starling Forces

- Describes fluid movement into (absorption) or out of (filtration) the capillary

Hydrostatic Pressure

- The primary force driving fluid transport between the capillaries and tissues is hydrostatic pressure, which can be defined as the pressure of any fluid enclosed in a space.

Oncotic Pressure

- Net pressure that drives reabsorption—the movement of fluid from the interstitial fluid back into the capillaries—is called osmotic pressure (sometimes referred to as oncotic pressure).

Whereas hydrostatic pressure forces fluid out of the capillary, osmotic pressure draws fluid back in.

Table 28-3 Hormones That Regulate Tubular Reabsorption

Hormone	Site of Action	Effects
Aldosterone	Collecting tubule and duct	↑NaCl, H ₂ O reabsorption; ↑K ⁺ secretion; ↑H ⁺ secretion
Angiotensin II	Proximal tubule, thick ascending loop of Henle, distal tubule, collecting tubule	↑NaCl, H ₂ O reabsorption; ↑H ⁺ secretion
Antidiuretic hormone	Distal tubule/collecting tubule and duct	↑H ₂ O reabsorption
Atrial natriuretic peptide	Distal tubule/collecting tubule and duct	↓NaCl reabsorption
Parathyroid hormone	Proximal tubule, thick ascending loop of Henle, distal tubule	↓PO ₄ ³⁻ reabsorption; ↑Ca ²⁺ reabsorption

ACID-BASE REGULATION

Regulation of Hydrogen Ion (H⁺) Balance

- Homeostasis requires balance between H⁺ intake/production and H⁺ removal.
- The kidneys play a key role in removing H⁺ from the body.
- Precise H⁺ control involves not just the kidneys but also:
 - **Buffer systems** in blood and cells
 - Respiratory regulation via the lungs
- **Bicarbonate ions** (HCO₃⁻) are crucial in acid-base regulation — the kidneys reabsorb, secrete, and excrete H⁺ and HCO₃⁻ to maintain balance.

Importance of Precise H⁺ Regulation

- H⁺ concentration affects enzyme activity, influencing almost all cellular functions.
- The normal extracellular H⁺ concentration is extremely low:
 - ≈ 0.00004 mEq/L (or 40 nEq/L)
 - 3.5 million times lower than sodium concentration.

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- Even tiny changes in H^+ levels can disrupt physiological processes — highlighting the importance of tight regulation.

Acid and Bases - Key Definitions

- **Acid:** Molecule that releases H^+ in solution.
 - Example: $HCl \rightarrow H^+ + Cl^-$
 - Carbonic acid (H_2CO_3) $\rightarrow H^+ + HCO_3^-$
- **Base:** Molecule or ion that accepts H^+ .
 - Examples: $HCO_3^- \rightarrow H_2CO_3$; $HPO_4^{2-} \rightarrow H_2PO_4^-$
- Proteins, including hemoglobin, act as important bases due to negatively charged amino acids.

Alkalinity and Terminology

- Base and alkali are often used synonymously.
 - Alkalis are combinations of alkaline metals (Na^+ , K^+) with OH^- .
- **Alkalosis:** Excess removal of H^+ from body fluids.
- **Acidosis:** Excess addition of H^+ to body fluids.

Strong vs. Weak Acids and Bases

- **Strong acids** (e.g., HCl): Fully dissociate $\rightarrow$ release large amounts of H^+ .
- **Weak acids** (e.g., H_2CO_3): Partially dissociate $\rightarrow$ release fewer H^+ .
- **Strong bases** (e.g., OH^-): Rapidly remove H^+ to form water.
- **Weak bases** (e.g., HCO_3^-): Bind H^+ weakly.
- Most physiological buffers (H_2CO_3/HCO_3^-) are weak acids and bases.

Normal H^+ Concentration and pH

- Normal $[H^+] = 40 \text{ nEq/L} \rightarrow \text{pH} = 7.4$
 - $\text{pH} = -\log[H^+]$; inversely related to H^+ concentration.
- **Arterial blood pH:** 7.4
- **Venous/interstitial pH:** 7.35 (due to $CO_2 \rightarrow H_2CO_3$ formation)
- **Intracellular pH:** 6.0–7.4 (slightly lower due to metabolic acids)
- **Urine pH:** 4.5–8.0 (varies with acid–base status)
- **Life-compatible range:** pH 6.8–8.0

- Below 6.8 $\rightarrow$ severe acidemia

- Above 8.0 $\rightarrow$ severe alkalemia

Stomach Acid

- Parietal (oxyntic) cells in the stomach secrete HCl with $\text{pH} \approx 0.8$.
- $[H^+]$ here is ~4 million times higher than in blood.
- Demonstrates the body's ability to create extreme local acidity while maintaining systemic pH stability.

Three Primary Systems Regulating H^+ Concentration

Chemical Buffer Systems (Immediate response)

- **First line of defense.**
 - Act within seconds to prevent major pH changes.
 - Function: Combine with acids or bases to temporarily neutralize H^+ changes.
 - Important note: Buffers do not eliminate or add H^+ — they only stabilize until balance is restored.

Respiratory System (Short-term response)

- **Second line of defense.**
 - Acts within minutes through the respiratory center in the brain
 - Regulates CO_2 removal, which in turn controls carbonic acid (H_2CO_3) levels.
 - $\uparrow CO_2 \rightarrow \uparrow H_2CO_3 \rightarrow \downarrow \text{pH}$ (acidosis)
 $\downarrow CO_2 \rightarrow \downarrow H_2CO_3 \rightarrow \uparrow \text{pH}$ (alkalosis)

Kidneys (Long-term response)

- **Third line of defense.**
 - Most powerful acid–base regulator, though slow-acting (hours to days).
 - Adjusts H^+ balance by excreting acidic or alkaline urine.
 - Restores extracellular fluid pH to normal during acidosis or alkalosis.

Defense System	Response Time	Function	Strength
Chemical Buffers	Seconds	Immediate neutralization of H^+	Rapid, temporary
Respiratory System	Minutes	Removes CO_2 to adjust H_2CO_3	Moderate
Kidneys	Hours–Days	Excretes acids/bases via urine	Slow but powerful

Definition and Basic Mechanism of Buffers

- A **buffer** is any substance that can reversibly bind H^+ to resist changes in pH.
 - General reaction: $\text{Buffer} + H^+ \leftrightarrow \text{H Buffer}$
- When $[H^+]$ increases, the reaction shifts to the right $\rightarrow$ more H^+ binds to the buffer.

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- When $[H^+]$ decreases, the reaction shifts to the left $\rightarrow H^+$ is released from the buffer.
- This reversible action allows buffers to stabilize pH and minimize sudden changes in H^+ concentration.

Significance of Buffer Systems

- Buffers prevent drastic pH shifts despite continuous acid production in the body.
- The body produces or ingests ~ 80 mEq of H^+ daily, mainly from metabolic processes.
- In contrast, the normal extracellular H^+ concentration is only 0.00004 mEq/L — extremely small.
- **Without buffering systems**, daily acid production would cause lethal changes in pH.

How Buffers Work

- Buffers do not eliminate acids or bases; they temporarily neutralize excess H^+ until the lungs and kidneys restore balance.
- They maintain acid–base stability by binding or releasing H^+ as needed.

Most Important Buffer System

- The **bicarbonate buffer system** (H_2CO_3 / HCO_3^-) is the primary and most effective buffer in extracellular fluid (ECF).
- It plays a major quantitative role in maintaining blood and ECF pH.

Bicarbonate Buffer System

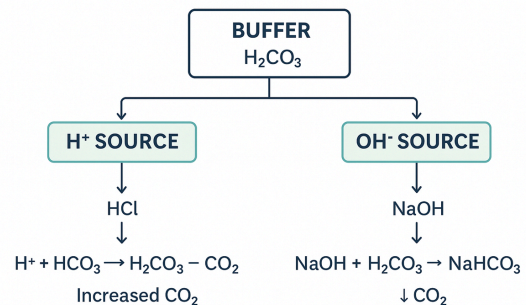
- Components: The system is a solution containing two crucial ingredients:
 - **A weak acid:** Carbonic Acid (H_2CO_3)
 - **A bicarbonate salt** (acting as a weak base): Bicarbonate (HCO_3^-), usually from Sodium Bicarbonate ($NaHCO_3$).
- **Formation of Carbonic Acid:** Carbonic acid is formed when carbon dioxide (CO_2) reacts with water (H_2O).
- This reaction is very slow on its own but is rapidly sped up by the enzyme carbonic anhydrase, which is found in lung alveoli and kidney tubule cells.
- $CO_2 + H_2O \leftrightarrow H_2CO_3$

- **The Central Equilibrium:** The entire system exists in a dynamic balance represented by the following equation.
- Because H_2CO_3 is a weak acid, the concentration of free H^+ is kept very low.
- $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$
- **How It Buffers Strong Acids:** When a strong acid (like HCl) adds H^+ to the blood, the bicarbonate ion (HCO_3^-) acts as a weak base, accepting the excess H^+ to form carbonic acid.
- $\uparrow H^+ + HCO_3^- \rightarrow H_2CO_3 \rightarrow CO_2 + H_2O$
- The carbonic acid then converts to carbon dioxide and water.
- The excess CO_2 stimulates respiration, and the lungs exhale it, effectively removing the acid from the body.
- **How It Buffers Strong Bases:** When a strong base (like NaOH) is added, its hydroxide ions (OH^-) are neutralized by the carbonic acid (H_2CO_3), a weak acid.
- $OH^- + H_2CO_3 \rightarrow HCO_3^- + H_2O$
- This reaction consumes H_2CO_3 , causing CO_2 levels to drop (inhibiting respiration) and HCO_3^- levels to rise.
- The excess HCO_3^- is then excreted by the kidneys to restore balance.

Integration with Lungs and Kidneys

- The lungs control CO_2 levels, influencing H_2CO_3 concentration.
- The kidneys regulate HCO_3^- reabsorption and excretion, ensuring long-term stability.
- Together, they keep blood pH near 7.4, preventing acidosis or alkalosis.

BICARBONATE BUFFER SYSTEM



Phosphate Buffer System

Key Components: This buffer system consists of

- a weak acid, dihydrogen phosphate ($H_2PO_4^-$), and
- a weak base, hydrogen phosphate (HPO_4^{2-}).

Against Strong Acid (e.g., HCl):

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- The base component, HPO_4^{2-} , accepts the excess hydrogen ion (H^+), converting the strong acid into a weak one (H_2PO_4^-).
- This minimizes the drop in pH.
- $\text{H}^+ + \text{HPO}_4^{2-} \rightarrow \text{H}_2\text{PO}_4^-$
- The acid component, H_2PO_4^- , donates a hydrogen ion to neutralize the hydroxide ion (OH^-), forming the weak base HPO_4^{2-} and water.
- This prevents a sharp rise in pH.
- $\text{OH}^- + \text{H}_2\text{PO}_4^- \rightarrow \text{HPO}_4^{2-} + \text{H}_2\text{O}$
- Limited Role in Extracellular Fluid (ECF): Although its optimal buffering pH ($\text{pK} = 6.8$) is close to the body's normal pH of 7.4, its concentration in the ECF is very low.
- It has only about 8% of the buffering power of the bicarbonate system in the blood.
- Crucial Role in Kidneys and Cells: The phosphate buffer system is vitally important in two specific areas:
 - Renal tubular fluid
 - Intracellular fluid
- Plays a major role in renal tubular fluid because:
- Phosphate becomes concentrated in tubules $\rightarrow$ enhances buffering capacity.
- Tubular fluid has lower pH, closer to phosphate's pK (6.8) $\rightarrow$ optimal buffering action.
- In Intracellular Fluid
- Phosphate concentration is much higher inside cells than in plasma.
- Intracellular pH is slightly lower $\rightarrow$ closer to pK , allowing efficient buffering of cellular acids and bases.

Key Concepts

- H^+ concentration must be tightly regulated — small deviations cause major physiological effects.
- Kidneys, lungs, and buffer systems work together to maintain pH homeostasis.
- Bicarbonate (HCO_3^-) and carbonic acid (H_2CO_3) are the **primary buffering system**.
- Acidosis and alkalosis describe processes leading to acidemia (low pH) and alkalemia (high pH), respectively.